

Advanced Self-Supervised Learning and JEPA Models

Naeemullah Khan

naeemullah.khan@kaust.edu.sa



جامعة الملك عبد الله
للعلوم والتقنية
King Abdullah University of
Science and Technology

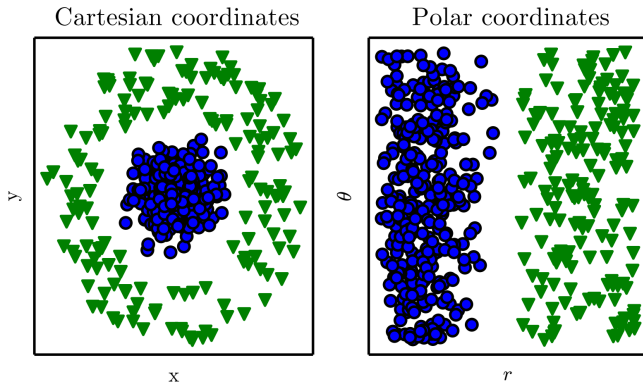
KAUST Academy
King Abdullah University of Science and Technology

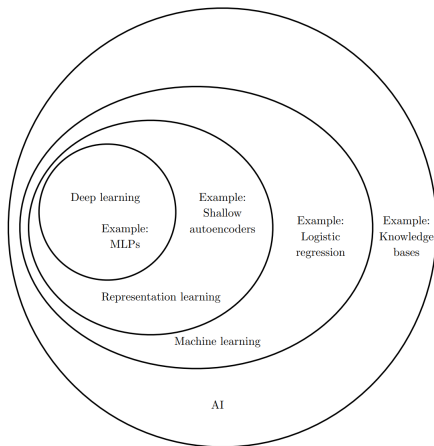
May 27, 2026

1. Introduction: What is Self-Supervised Learning (SSL)?
2. Evolution of SSL and a Three-Way Taxonomy (invariant, generative, JEPA)
3. Theoretical foundations and why SSL transfers
4. JEPA motivation: world models and objective-driven AI
5. I-JEPA: paradigms, architecture, training, and efficiency
6. From images to video: V-JEPA (masking, probes); **V-JEPA 2** (scale, LLM alignment, planning)
7. Specialized extensions: Point-JEPA, CNN-JEPA, VL-JEPA; video efficiency (SALT)
8. Open problems, further study (courses & surveys), and references

SSL: Introduction

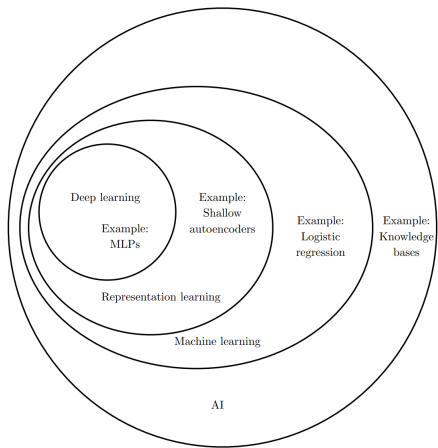
Representations Matter





Deep Unsupervised Learning:

- ▶ Learns data representations without using labels.
- ▶ Is a subset of Deep Learning, which itself is a subset of Representation Learning, all within Machine Learning.



Deep Unsupervised Learning:

- ▶ Learns data representations without using labels.
- ▶ Is a subset of Deep Learning, which itself is a subset of Representation Learning, all within Machine Learning.

Self-Supervised Learning (SSL):

- ▶ A form of unsupervised learning that creates supervision from the data itself by designing *pretext tasks*.
- ▶ These tasks generate pseudo-labels, enabling the model to learn useful representations without manual annotation.
- ▶ Often used interchangeably with unsupervised learning, but SSL specifically focuses on leveraging intrinsic data structure.

SSL is a type of unsupervised learning where the data itself provides supervision. Typically, part of the data is hidden, and a neural network is trained to predict it from the rest—this defines the *pretext task*. Well-chosen pretext tasks help models learn useful features without manual labels.

- ▶ **Pretext Task:** Examples include predicting masked patches, future representations (CPC), or distinguishing positive/negative pairs (contrastive).
- ▶ **Downstream Task:** After pre-training, the encoder is fine-tuned for tasks like classification or detection.
- ▶ **Key Trade-Offs:**
 - **Reconstructive** (e.g., autoencoders): capture low-level details, may miss semantics.
 - **Predictive/Contrastive** (e.g., CPC, MoCo): focus on high-level information.

What is Self-Supervised Learning? (cont.)

► “Pure” Reinforcement Learning (**cherry**)

- The machine predicts a scalar reward given once in a while.
- **A few bits for some samples**

► Supervised Learning (**icing**)

- The machine predicts a category or a few numbers for each input
- Predicting human-supplied data
- **10→10,000 bits per sample**

► Self-Supervised Learning (**cake génoise**)

- The machine predicts any part of its input for any observed part.
- Predicts future frames in videos
- **Millions of bits per sample**



Why Self-Supervised Learning?



“Give a robot a label and you feed it for a moment; teach a robot to label and you feed it for a lifetime.”

— Pierre Sermanet

- ▶ **High cost of dataset creation:** Each new task often requires building a new labeled dataset.
 - Involves preparing labeling manuals, defining categories, hiring annotators, building GUIs, and managing storage pipelines.
- ▶ **Expensive supervision:** High-quality labels can be costly or impractical to obtain (e.g., in medicine or legal domains).
- ▶ **Leverage unlabeled data:** The internet provides vast amounts of unlabeled images, videos, and text that can be utilized.
- ▶ **Cognitive motivation:** Animals and babies learn from their environment without explicit supervision, inspiring SSL approaches.

- ▶ **Early methods:** rotation prediction, jigsaw, colorization—cheap supervision, limited semantic depth.
- ▶ **Contrastive SSL:** SimCLR, MoCo, BYOL, VICReg—align embeddings across views while avoiding collapse.
- ▶ **Masked modeling / generative SSL:** MAE, BEiT—reconstruct tokens or pixels in observation space.
- ▶ **Predictive in latent space (JEPA):** I-JEPA, V-JEPA—predict *representations* of held-out regions, not raw observations.

Cross-method survey of SSL for ViTs: [arXiv:2408.17059](https://arxiv.org/abs/2408.17059).

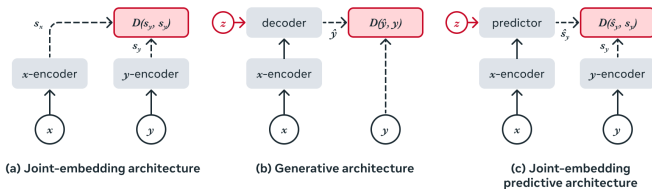
Three Architectural Families in Self-Supervised Learning

Family	Core mechanism	Representative examples
Joint-embedding (<i>invariant</i>)	Pull together compatible views; contrastive or regularized losses prevent collapse.	SimCLR, MoCo, BYOL, VICReg
Generative (<i>reconstruction</i>)	Fill in missing input in pixel or token space; model must explain fine observation detail.	MAE, BEiT, VAEs, GAN-style objectives
JEPA (<i>latent prediction</i>)	Predict embedding of a target region from context embeddings—semantics without full pixel likelihood.	I-JEPA, V-JEPA, V-JEPA 2, Point-JEPA, CNN-JEPA

- ▶ Exploits **structure** in natural data (spatial context, temporal continuity, language statistics).
- ▶ Contrastive methods stress **invariances** to augmentations; reconstruction methods force **complete** explanations of the input; JEPA emphasizes **predictive** structure in **representation space**.
- ▶ Strong **linear / low-data probes** suggest encoders expose semantically useful axes for downstream tasks.
- ▶ Active research: theoretical limits, failure modes (shortcuts), and unified views across the three families above.

SSL Paradigms: Invariant, Generative, and Predictive (JEPA)

- ▶ (a) **Joint-embedding (invariance)**: map views x, y to embeddings and pull compatible pairs together—can bias toward invariance shortcuts.
- ▶ (b) **Generative**: decode $\hat{y}$ from context x (and latent z), loss in **pixel/input space**—must explain unpredictable detail.
- ▶ (c) **JEPA**: predict **embedding** $\hat{s}_y$ of y from context; compare to s_y from a target encoder—semantic, discard irrelevant variability.



Reference: Adapted from Meta I-JEPA blog.

- ▶ Most self-supervised learning (SSL) approaches focus on predicting local features or reconstructing pixels. Can we go further and predict high-level, abstract embeddings instead?
- ▶ **JEPA's key idea:** Predict the *representations* of masked or missing regions, not just raw pixel values. This approach enables semantically rich, meaningful representations.
- ▶ JEPA and world models are proposed as a path to machines that internalize how the world works. By predicting abstract representations rather than details, models can focus on semantics and ignore unpredictable noise.
- ▶ This shift reflects a broader vision (Y. LeCun et al.): powerful architectures like Transformers, while impressive, still lack the human/animal ability to acquire deep understanding from little labeled data. Purely scaling models does not guarantee human-like world understanding (**Moravec paradox**).

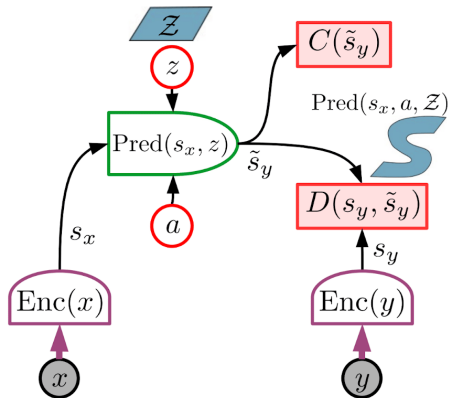
What are Joint Embedding Predictive Architectures (JEPA)?

Models learn by predicting the **embedding** of one part of the input (e.g. a future segment or a masked region) from the representation of another (context / past), using paired encoders and a predictor. For example, predict the embedding of a masked image region (target) from the visible (context) regions.

- ▶ Do not reconstruct pixels; learn on the level of abstract representations.
- ▶ The primary training signal is in **representation space**.
- ▶ Concrete models include I-JEPA (images), V-JEPA (video), MC-JEPA (motion/content), and extensions to other modalities.

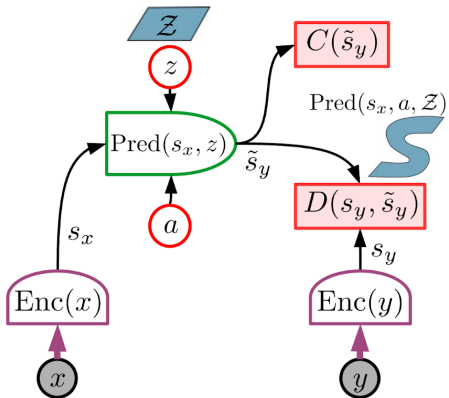
► JEPA: Joint Embedding Predictive Architecture.

- x : observed past and present
- y : future
- a : action
- z : latent variable (unknown)
- $D()$: prediction cost
- $C()$: surrogate cost



Reference: Adapted from Turing Post / Meta AI core JEPA diagram.

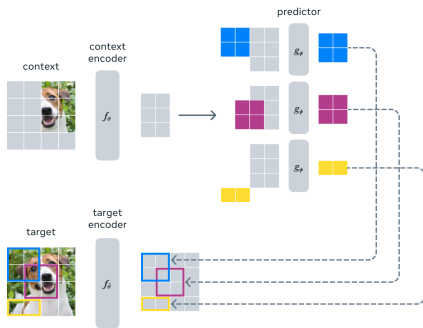
- ▶ Observed x (past/present) and y (future or target) map to representations s_x, s_y .
- ▶ A **predictor** maps (s_x, a, z) to $\tilde{s}_y$; optional **action** a and latent z capture aspects of y not fixed by x .
- ▶ Training minimizes a prediction cost $D(s_y, \tilde{s}_y)$.



Reference: Adapted from Turing Post / Meta AI core JEPA diagram.

- ▶ Introduced by Meta AI (Assran et al., 2023; ICLR 2023) as an instantiation of Yann LeCun's **Joint Embedding Predictive Architecture** vision: learn an *internal model* of the world by comparing **abstract representations** rather than raw pixels.
- ▶ Self-supervised from unlabeled images; targets are **feature embeddings** of masked regions predicted from visible context, not pixel reconstruction.

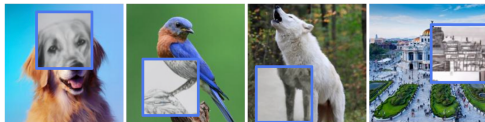
- ▶ Generative objectives often try to explain **every** missing detail even when the world is **stochastic** at pixel level—wasting capacity on irrelevant variation (Meta blog example: hands in generative models).
- ▶ Predicting **embeddings** steers the model toward **stable, high-level** structure: objects, parts, and layout.
- ▶ Empirically: strong **linear probing** and **semi-supervised** transfer; competitive on some **low-level** tasks (e.g. counting, depth) despite simpler augmentation than many contrastive methods.



Reference: Meta I-JEPA blog architecture.

- ▶ **Context encoder (f_θ):** ViT on **visible** patches only.
- ▶ **Predictor (g_ϕ):** narrow ViT that maps context tokens + target **positional** information to predicted embeddings for each target block.
- ▶ **Target encoder (f_ϕ):** encodes full image for targets; weights track the context encoder via **exponential moving average (EMA)**, not gradient through targets.
- ▶ Training minimizes distance between predicted and target **embeddings**.

- ▶ To **visualize** what the predictor represents, Meta trains a **stochastic decoder** that maps predicted embeddings toward **pixel sketches** inside a blue box (for analysis only—not the training objective).
- ▶ The model tends to capture **object parts** and **pose** (e.g. dog's head, wolf legs, building structure) rather than arbitrary texture noise.



Illustrating how the predictor learns to model the semantics of the world. For each image, the portion outside of the blue box is encoded and given to the predictor as context. The predictor outputs a representation for what it expects to be in the region within the blue box. To visualize the prediction, we train a generative model that produces a sketch of the contents represented by the predictor output, and we show a sample output within the blue box. Clearly the predictor recognizes the semantics of what parts should be filled in (the top of the dog's head, the bird's leg, the wolf's legs, the other side of the building).

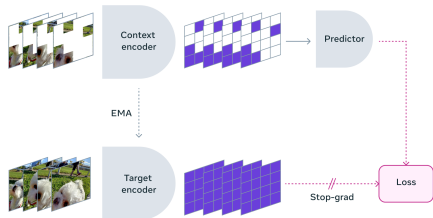
Method	Architecture	Epochs	Top-1 accuracy
<i>Methods without view data augmentations</i>			
data2vec	ViT-L/16	1600	73.3
MAE	ViT-L/16	1600	67.1
	ViT-H/14	1600	71.5
I-JEPA	ViT-L/16	600	69.4
	ViT-H/14	300	73.3
	ViT-H/16 ₄₄₈	300	77.3
<i>Methods using extra view data augmentations</i>			
iBOT	ViT-B/16	250	69.7
DINO	ViT-B/8	300	70.0
SimCLR v2	RN151 (2x)	800	70.2
BYOL	RN200 (2x)	800	71.2
MSN	ViT-B/4	300	75.7

Low Shot Classification Accuracy: Semi-supervised evaluation on ImageNet-1k with 1% of the labels (approximately 12 labeled images per class).

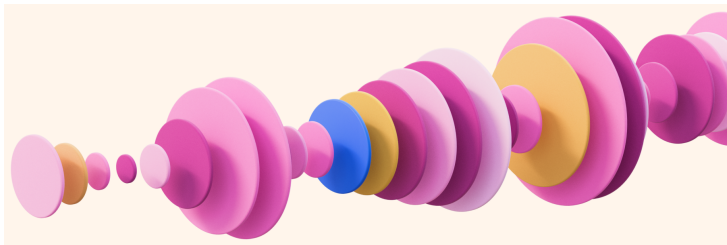
- ▶ Strong **off-the-shelf** representations: linear probe, semi-supervised, and competitive low-shot settings without extensive task-specific fine-tuning of the backbone.
- ▶ Transfers to classification, detection, segmentation-style use cases; blog and paper emphasize **broader task mix** including some low-level vision tasks.
- ▶ Open direction: extend JEPAs to **richer modalities** (video, audio, text conditioning) for longer-horizon **world models**.

- ▶ The visual world is dynamic: understanding actions and events requires temporal structure.
- ▶ **V-JEPA** (Video Joint Embedding Predictive Architecture, Meta) extends the JEPA idea to video: predict *representations* of masked spatio-temporal content, not pixels.
- ▶ Like I-JEPA, it is **non-generative**: learning targets live in a learned latent space, so the model can ignore unpredictable detail.
- ▶ Pre-training uses **unlabeled video**; labeled data is reserved for lightweight adaptation.

- ▶ Encoder trained by predicting masked spatio-temporal regions in a **learned latent space**—analogous to I-JEPA comparing abstract image embeddings rather than pixels.
- ▶ Self-supervised pre-training on diverse video; task labels used only after pre-training.



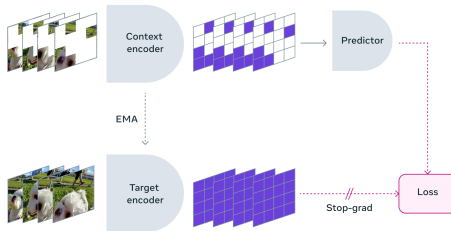
Encoder trained by predicting masked regions in latent space (not pixels). Reference: [V-JEPA paper \(Baevski et al., 2023\)](#).



Spatio-temporal context and masking. Reference: Meta V-JEPA blog.

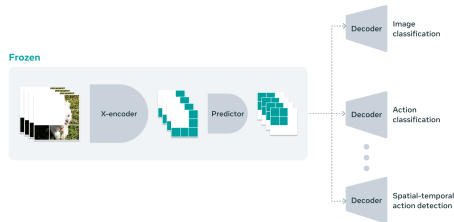
- ▶ Masking must be hard enough: **sparse random patches** make the task too easy—little is learned about dynamics.
- ▶ If a region is masked only at **one instant** while adjacent frames stay visible, the model can “cheat” from neighboring time steps.
- ▶ V-JEPA uses masking across **both space and time** so the model must infer what happens in the scene, not copy texture from nearby frames.

- ▶ **Context path:** encodes visible video patches; **predictor** maps to the target representation space (JEPA-style).
- ▶ **Target path:** encodes masked regions for the training target (with stop-gradient); loss matches predicted and target embeddings.
- ▶ Same high-level recipe as I-JEPA, extended to **3D** patch tokens over space and time.



Reference: [V-JEPA paper \(Baevski et al., 2023\)](#).

- ▶ After self-supervised pre-training, the **encoder and predictor are frozen**.
- ▶ Same frozen backbone can support several tasks (e.g., action classification, fine-grained interaction recognition, activity localization) without full fine-tuning that would specialize the whole network to one benchmark.



Reusing frozen representations for downstream video understanding. Reference: **V-JEPA paper (Baevski et al., 2023)**.

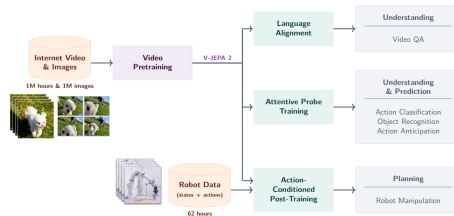
- ▶ Strong on **frozen** benchmarks spanning image/video classification, action recognition, and spatio-temporal action detection relative to prior video representation learning.
- ▶ Particularly strong on **fine-grained** interaction understanding over short clips (e.g., distinguishing put-down vs. pick-up vs. pretend actions).
- ▶ Reported limitations include **short clip** focus (on the order of seconds); **audio** and **long-horizon** prediction/planning highlighted as future directions.

Method	Arch.	Frozen Evaluation					
		K400 (16×8×3)			SSv2 (16×2×3)		
		5%	10%	50%	5%	10%	50%
MVD	ViT-L/16	62.6 ± 0.2	68.3 ± 0.2	77.2 ± 0.3	42.9 ± 0.8	49.5 ± 0.6	61.0 ± 0.2
VideoMAE	ViT-H/16	62.3 ± 0.3	68.5 ± 0.2	78.2 ± 0.1	41.4 ± 0.8	48.1 ± 0.2	60.5 ± 0.4
VideoMAEv2	ViT-g/14	37.0 ± 0.3	48.8 ± 0.4	67.8 ± 0.1	28.0 ± 1.0	37.3 ± 0.3	54.0 ± 0.3
V-JEPA	ViT-H/16 ₃₈₄	68.2 ± 0.2	72.8 ± 0.2	80.6 ± 0.2	54.0 ± 0.2	59.3 ± 0.5	67.9 ± 0.2

V-JEPA 2: Scaling Video JEPA for Understanding and Planning

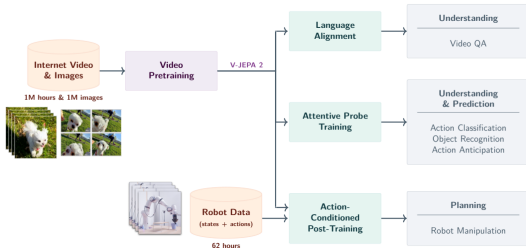
Same **visual mask denoising** / JEPA spirit as I-JEPA, but at **internet scale**

- ▶ Order of **~1M hours** of video and **~1M images** for pre-training.
- ▶ Downstream use: video understanding, video QA, and planning.

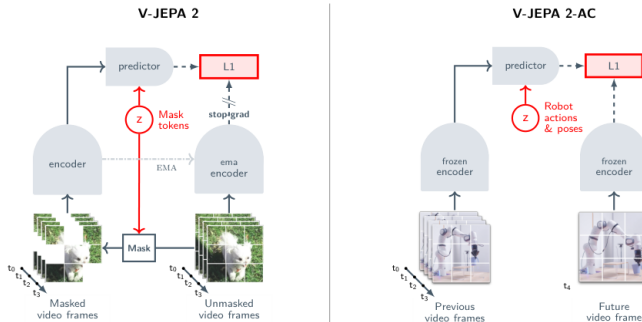


Assran et al. (2025): overview of pre-training, alignment, and robotics post-training.

- ▶ **Stage A:** SSL on web video/images → strong motion and appearance understanding.
- ▶ **Stage B: language alignment**—project video tokens into an LLM with small attentive stacks so the backbone supports VQA and reasoning.
- ▶ **Stage C:** optional **action-conditioned** head (V-JEPA 2-AC) on tens of hours of robot data for latent planning.



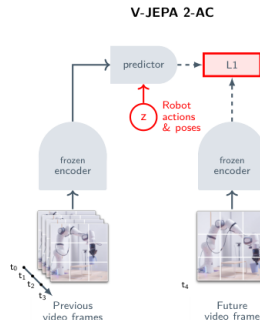
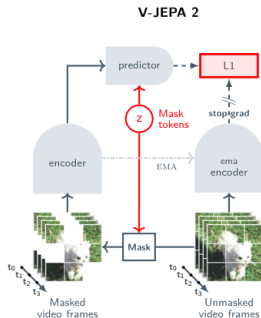
Overview of pre-training, alignment, and robotics post-training.



Assran et al. (2025): multistage training.

► V-JEPA 2:

- Video clips divided into tokens; a subset is masked.
- Encoder processes masked sequence to produce token embeddings.
- Mask tokens concatenated with encoder outputs.
- Predictor regresses to targets computed by an EMA encoder using L_1 loss.



Assran et al. (2025): multistage training.

► V-JEPA 2-AC:

- Freeze pretrained encoder and train action-conditioned predictor on top.
- Use autoregressive prediction to model future frame representations based on past frames, actions, and end-effector states.

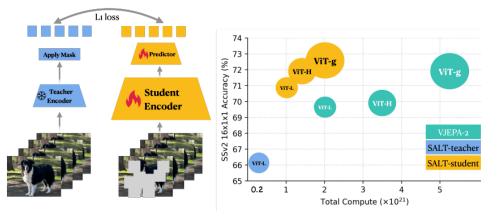
TASK TYPE	BENCHMARK	V-JEPA 2	PREVIOUS BEST
Planning and Robot Control from Image Goals	Reach	100%	100% (Octo)
	Grasp	45%	8% (Octo)
	Pick-and-place	73%	13% (Octo)
Prediction	Epic-Kitchens-100 action anticipation	39.7%	27.6% (PlausiVL)
Understanding	Something-Somethingv2 action recognition Attentive probe	77.3%	69.7% (InternVideo2-1B Attentive probe)
	Diving48 Attentive probe	90.2%	86.4% (InternVideo2-1B Attentive probe)
	Perception Test	84.0%	82.7% (PerceptionLM)
	MVPBench	44.5%	39.9% (InternVL-2.5)

Assran et al. (2025): performance; see also [Meta AI V-JEPA blog post](#).

- ▶ **Point-JEPA:** applies the JEPA recipe to **unordered 3D point sets**; uses spatial proximity and masking in the point domain so the model predicts **latent structure** of masked regions from geometric context (not pixels).
- ▶ **CNN-JEPA:** keeps **convolutional** encoders (sparse, translation-friendly) while retaining **latent prediction** rather than pixel reconstruction—useful when ViTs are not the deployment backbone.
- ▶ Related 3D SSL: **Sonata** (2025) pursues highly reliable point representations from large-scale self-supervision—complementary reading for geometry-heavy applications.
- ▶ **VL-JEPA** extends the same **predictive latent** philosophy to **vision–language** setups: build world models that couple visual structure with language for richer situational understanding.

- ▶ **SALT** (*Static-teacher Asymmetric Latent Training*, Li et al., 2025) uses a **two-stage recipe**: (i) train a teacher with a **pixel reconstruction** objective under V-JEPA-style masking; (ii) **freeze** the teacher and train a student to predict teacher latents on masked regions.
- ▶ Reported benefits: simpler optimization than coupled EMA schedules, favorable **accuracy–compute** trade-offs in frozen evaluation, and student quality surprisingly robust to teacher size.

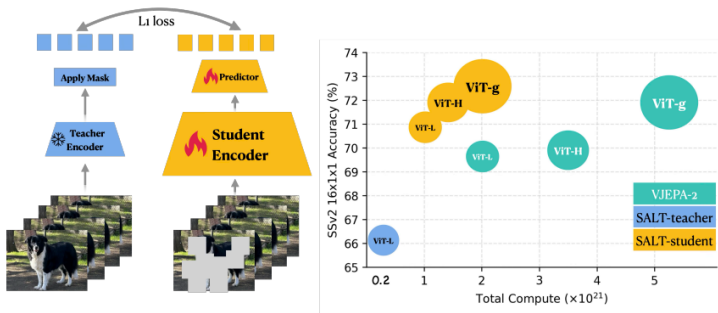
SALT: Two-Stage Recipe for Video SSL (Stage 1)



Source: Li et al. (2025) SALT, [arXiv:2509.24317](https://arxiv.org/abs/2509.24317), Fig. 2.

- **Goal:** Train a **teacher encoder** to robustly reconstruct masked video regions *at the pixel level*.
- **Approach:** Use a **V-JEPa-style masking** scheme; teach the encoder to infill missing pixels based on visible context.
- **Outcome:** Teacher encoder learns to summarize temporal and spatial structure, producing feature-rich latent representations.

SALT: Two-Stage Recipe for Video SSL (Stage 2)



Reference: Li et al. (2025) SALT, [arXiv:2509.24317](https://arxiv.org/abs/2509.24317), Fig. 2.

- **Freeze** the teacher encoder weights.
- Train a student to **predict teacher latent embeddings** of masked regions, using only visible context as input.
- Mimics standard JEPa: loss is in **latent space**, not directly on pixels.

- ▶ **JEPA: A new paradigm for Self-Supervised Learning**
Joint Embedding Predictive Architectures (JEPA) learn *internal world models* by predicting **abstract representations** of masked or future regions, instead of raw data, enabling robust and efficient learning.
- ▶ **Contrast to Classic SSL:** JEPA unifies ideas from contrastive (invariant) and generative SSL, but avoids pixel-level prediction.
- ▶ **Strong Transfer:** Predicting stable, high-level embedding targets yields representations that excel at downstream tasks such as VQA, reasoning, and planning.
- ▶ **Flexible Recipe:**
The JEPA framework extends to images (I-JEPA), video (V-JEPA, V-JEPA 2), point clouds (Point-JEPA), vision transformers and CNNs (CNN-JEPA), and multimodal data (VL-JEPA).
- ▶ **Active Areas:** Ongoing work explores efficient teachers (SALT), hierarchical and recurrent world models, multimodal fusion, and theoretical understanding of latent prediction.

- ▶ **Masking and targets:** How much to mask, where, and at what semantic scale (patch vs. object vs. clip)?
- ▶ **Long horizon:** Original V-JEPA focused on short clips; **V-JEPA 2** pushes scale and adds action-conditioned planning (V-JEPA 2-AC), but long-horizon world models remain open.
- ▶ **Multimodal fusion:** aligning audio, language, and video under JEPA-style latent prediction without collapsing modalities.
- ▶ **Theory:** when does latent prediction provably recover predictable structure, and when do shortcuts dominate?
- ▶ **Deployment:** latency, memory, and hardware-aware training for billion-scale video corpora.

1. Assran et al., “Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture,” ICLR 2023. [arXiv:2301.08243](#)
2. Bardes et al., “Revisiting Feature Prediction for Learning Visual Representations from Video,” 2024. [arXiv:2404.08471](#)
3. LeCun, “A Path Towards Autonomous Machine Intelligence,” OpenReview 2022. [OpenReview](#)
4. Meta AI, “I-JEPA: The first AI model based on Yann LeCun’s vision for more human-like AI,” *AI at Meta* blog, 2023. [ai.meta.com](#)
5. Meta AI, “Yann LeCun on a vision to make AI systems learn and reason like animals and humans,” *AI at Meta* blog. [ai.meta.com](#)
6. Meta AI, “Introducing V-JEPA 2,” V-JEPA research hub. [ai.meta.com/research/vjepa](#)
7. Assran et al., “V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning,” 2025. [arXiv:2506.09985](#)
8. “Tutorial on Joint Embedding Predictive Architectures (JEPA),” TechRxiv. [DOI:10.36227/techrxiv.176469421.19270944](#)
9. Kalapos & Gyires-Tóth, “CNN-JEPA: Self-Supervised Pretraining Convolutional Neural Networks Using Joint Embedding Predictive Architecture,” 2024. [arXiv:2408.07514](#)

10. Saito et al., “Point-JEPA: A Joint Embedding Predictive Architecture for Self-Supervised Learning on Point Cloud,” 2024. [arXiv:2404.16432](#)
11. Wu et al., “Sonata: Self-Supervised Learning of Reliable Point Representations,” CVPR 2025. [arXiv:2503.16429](#)
12. Li et al., “Rethinking JEPA: Compute-Efficient Video SSL with Frozen Teachers” (SALT), Apple ML Research, 2025. [arXiv:2509.24317](#)
13. Khan et al., “A Survey of the Self Supervised Learning Mechanisms for Vision Transformers,” 2024. [arXiv:2408.17059](#)
14. Meta AI, “V-JEPA: The next step toward advanced machine intelligence,” *AI at Meta* blog, 2024. [ai.meta.com](#)
15. He et al., “Masked Autoencoders Are Scalable Vision Learners,” CVPR 2022.
16. Baevski et al., “data2vec: A General Framework for Self-supervised Learning in Speech, Vision and Language,” ICML 2022.
17. Chen et al., “A Simple Framework for Contrastive Learning of Visual Representations (SimCLR),” ICML 2020.
18. Devlin et al., “BERT,” NAACL 2019.
19. Touvron et al., “An Image is Worth 16x16 Words (ViT),” ICLR 2021.
20. Stanford CS231n, Lecture 12 slides (SSL). [PDF](#)

21. UC Berkeley CS294-158 (SP20), Deep Unsupervised Learning. [course site](#)
22. CMU 11-785 Deep Learning. [deeplearning.cs.cmu.edu](#)
23. MIT 6.S191 Introduction to Deep Learning. [introtodeeplearning.com](#)
24. Kuka, V., “What is JEPA?” *Turing Post*, 2024. [turingpost.com](#)
25. LeCun, Les Houches lecture slides (AMI / world models). [PDF](#)